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Pedal device

Abstract

A pedal device includes a pedal, a connecting rod, a reaction force generating mechanism, a housing having a housing hole for allowing the connecting rod to pass therethrough when the pedal is stepped on by an operator, and a cover member. The cover member includes a first connecting portion and a covering portion. The first connecting portion is connected to a housing end portion of the housing forming a housing hole. The covering portion is arranged at a position between the first connecting portion and a back surface of the pedal and is connected to the first connecting portion to cover the housing hole.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) The present application is a continuation application of International Patent Application No. PCT/JP2022/013792 filed on Mar. 24, 2022, which designated the U.S. and claims the benefit of priority from Japanese Patent Application No. 2021-082794 filed on May 14, 2021. The entire disclosures of all of the above applications are incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates to a pedal device.

BACKGROUND

(2) Conventionally, an automobile pedal provided with a covering member that covers a pedal stepped on by a driver of the vehicle is known.

SUMMARY

(3) According to an aspect of the present disclosure, a pedal device includes: a pedal configured to rotate about a rotation axis when being stepped on by an operator; a connecting rod connected to a back surface of the pedal, opposite to a stepping surface stepped on by the operator, to rotate with the pedal about the rotation axis; a reaction force generating mechanism configured to generate a reaction force against a stepping force of the operator applied to the pedal by receiving a force from the pedal through the connecting rod and by an elastic deformation when the pedal is stepped on by the operator; a housing having a housing hole through which the connecting rod passes when the pedal is stepped on by the operator, and accommodating the reaction force generating mechanism; and a cover member. The cover member includes a housing connection portion and a covering portion. The housing connection portion may be connected to a housing end portion which is an end portion defining the housing hole in the housing. The covering portion may be arranged at a position between the housing connection portion and the back surface of the pedal, and may be connected to the housing connection portion, to cover the housing hole.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Objects, features, and advantages of the present disclosure will become more apparent from the following detailed description made with reference to the accompanying drawings, in which:
- (2) FIG. 1 is a configuration diagram of a brake-by-wire system using a pedal device of a first embodiment;
- (3) FIG. 2 is a perspective view of the pedal device;
- (4) FIG. 3 is another perspective view of the pedal device;
- (5) FIG. 4 is a cross-sectional view of the pedal device;
- (6) FIG. 5 is a perspective view of a portion of the pedal device;
- (7) FIG. 6 is an enlarged view of a VI portion in FIG. 4;
- (8) FIG. 7 is a cross-sectional view of a connecting rod, a cover member, and a housing end portion of the pedal device;
- (9) FIG. 8 is a cross-sectional view of a pedal device of a second embodiment;
- (10) FIG. 9 is a cross-sectional view of a pedal device of a third embodiment;
- (11) FIG. 10 is a cross-sectional view of a pedal device of a fourth embodiment;
- (12) FIG. 11 is a cross-sectional view of a pedal device of a fifth embodiment;
- (13) FIG. 12 is a cross-sectional view of a connecting rod, a cover member, and a housing end of a pedal device of a sixth embodiment;
- (14) FIG. 13 is a cross-sectional view of a semiconductor device taken along a line XIII-XIII of FIG. 12;
- (15) FIG. 14 is a cross-sectional view of a housing and a cover member of a pedal device of a seventh embodiment;
- (16) FIG. 15 is a cross-sectional view of a housing and a cover member of a pedal device of an eighth embodiment;
- (17) FIG. 16 is a cross-sectional view of a housing and a cover member of a pedal device of a ninth embodiment;
- (18) FIG. 17 is a cross-sectional view of a connecting rod, a cover member, and a housing end of a pedal device of a tenth embodiment;
- (19) FIG. 18 is a cross-sectional view of a connecting rod, a cover member, and a housing end of a pedal device of an eleventh embodiment;
- (20) FIG. 19 is a cross-sectional view of a connecting rod, a cover member, and a housing end of a pedal device of a twelfth embodiment;
- (21) FIG. 20 is a cross-sectional view of a connecting rod, a cover member, and a housing end of a pedal device of a thirteenth embodiment;
- (22) FIG. 21 is a cross-sectional view of a connecting rod, a cover member, and a housing end of a pedal device of a fourteenth embodiment;
- (23) FIG. 22 is a cross-sectional view of a connecting rod, a cover member, and a housing end of a pedal device of a fifteenth embodiment;
- (24) FIG. 23 is a cross-sectional view of a connecting rod, a cover member, and a housing end of a pedal device of a sixteenth embodiment;
- (25) FIG. 24 is a cross-sectional view of a connecting rod, a cover member, and a housing end of a pedal device of a seventeenth embodiment;
- (26) FIG. 25 is a cross-sectional view of a connecting rod, a cover member, and a housing end of a pedal device of an eighteenth embodiment;
- (27) FIG. 26 is a cross-sectional view of a connecting rod, a cover member, and a housing end of a pedal device of a nineteenth embodiment;
- (28) FIG. 27 is a cross-sectional view of a connecting rod and a cover member of a pedal device of

a twentieth embodiment;

(29) FIG. **28** is a cross-sectional view of a connecting rod and a cover member of a pedal device of a twenty-first embodiment;

(30) FIG. **29** is a cross-sectional view of a first housing and a second housing of a pedal device of a twenty-second embodiment;

(31) FIG. **30** is a cross-sectional view of a first housing and a second housing of a pedal device of a twenty-third embodiment;

(32) FIG. **31** is a perspective view of a first housing of a pedal device of a twenty-fourth embodiment;

(33) FIG. **32** is a perspective view of a first housing of a pedal device of a twenty-fifth embodiment;

(34) FIG. **33** is a perspective view of a first housing of a pedal device of a twenty-sixth embodiment;

(35) FIG. **34** is a cross-sectional view of a pedal device of a twenty-seventh embodiment;

(36) FIG. **35** is a cross-sectional view of a pedal device of a twenty-eighth embodiment;

(37) FIG. **36** is a perspective view of a pedal device of a twenty-ninth embodiment;

(38) FIG. **37** is a side view of the pedal device;

(39) FIG. **38** is a side view of the pedal device;

(40) FIG. **39** is a side view showing a maximum depression state of the pedal device;

(41) FIG. **40** is a side view of a pedal device of a thirtieth embodiment;

(42) FIG. **41** is a cross-sectional view of a pedal, a connecting rod, a cover member, and a housing end portion of a pedal device of an another embodiment;

(43) FIG. **42** is a cross-sectional view of a pedal, a connecting rod, a cover member, and a housing end portion of a pedal device of an another embodiment;

(44) FIG. **43** is a cross-sectional view of a pedal, a connecting rod, a cover member, and a housing end portion of a pedal device of an another embodiment; and

(45) FIG. **44** is a cross-sectional view of a pedal, a connecting rod, a cover member, and a housing end portion of a pedal device of an another embodiment.

DETAILED DESCRIPTION

(46) According to studies by the inventors, when a surface of a pedal that is stepped on by a driver is simply covered with a covering member, the covering member may be worn or damaged by being stepped on by the driver. For this reason, foreign matter such as dust, moisture, and abrasion powder of the covering member may enter into an interior of a housing that accommodates a reaction force generating mechanism such as a spring through the damaged portion of the covering member.

(47) It is an object of the present disclosure to provide a pedal device that suppresses intrusion of foreign matter thereinto.

(48) According to an aspect of the present disclosure, a covering portion is arranged at a position between a housing connection portion and a back surface of a pedal. Thus, the covering portion cannot be stepped on by an operator. Therefore, the cover member is prevented from being worn and damaged. Further, since the covering portion covers a housing hole, foreign matter cannot enter into a housing through the housing hole from the outside of the housing. Thus, intrusion of foreign matter can be effectively restricted.

(49) Hereinafter, embodiments will be described with reference to the drawings. In the following embodiments, the same or equivalent portions are denoted by the same reference numerals, and the description thereof will be omitted.

First Embodiment

(50) A pedal device **1** of the present embodiment is used, for example, as a brake pedal in a brake-by-wire system **150** that controls the brakes of a vehicle. First, this brake-by-wire system **150** will be described.

(51) The brake-by-wire system **150** includes wheel cylinders **131** to **134**, an ECU **110**, a brake circuit **120** and a pedal device **1**, as shown in FIG. 1.

(52) The wheel cylinders **131** to **134** are arranged for each wheel of the vehicle. A brake pad (not shown) is attached to each of the wheel cylinders **131** to **134**.

(53) The ECU **110** has a first ECU **111** and a second ECU **112**. The first ECU **111** has a microcomputer, a drive circuit, or the like (not shown). The first ECU **111** controls a first brake circuit **121** of the brake circuit **120**, which will be described later, based on a signal from the pedal device **1**, which will be described later. The second ECU **112** has a microcomputer, a drive circuit, or the like (not shown). Further, the second ECU **112** controls a second brake circuit **122** of the brake circuit **120** described later based on a signal from the pedal device **1** described later.

(54) The brake circuit **120** includes the first brake circuit **121** and the second brake circuit **122**. The first brake circuit **121** includes a reservoir **124**, a motor **123**, a gear mechanism **125** and a master cylinder **126**. The reservoir **124** stores brake fluid. The motor **123** drives the gear mechanism **125**. The gear mechanism **125** reciprocates a master piston **127** of the master cylinder **126** in an axial direction of the master cylinder **126**. The second brake circuit **122** includes an electromagnetic valve (not shown) or the like. Further, the second brake circuit **122** controls the hydraulic pressure of each of the wheel cylinders **131** to **134** by opening and closing electromagnetic valves according to control signals from the second ECU **112**.

(55) Here, in order to describe the pedal device **1** below, the front-rear direction of the vehicle is defined as a front-rear direction D_a of the vehicle. The vertical direction of the vehicle is defined as a vertical direction d_b of the vehicle. The left-right direction of the vehicle is defined as a left-right direction D_c of the vehicle. Further, the front of the vehicle in the front-rear direction D_a may be referred to as a front part. The rear of the vehicle in the front-rear direction D_a may be referred to as a rear part. The upper side of the vehicle in the vertical direction d_b may be referred to as an upper part. The lower side of the vehicle in the vertical direction d_b may be referred to as a lower part. The left side of the vehicle in the left-right direction D_c may be referred to as a left part. The right side of the vehicle in the left-right direction D_c may be referred to as a right part.

(56) The pedal device **1** is an organ-type pedal device **1**. Therefore, in the pedal device **1**, a portion of the pedal **40** that is in front of the vehicle relative to a rotation axis CL rotates toward a floor **2** or toward a dash panel in a vehicle interior in response to an increase in a stepping force of a driver applied to the pedal **40**, which will be described later. The dash panel is a partition wall that separates the vehicle interior from the vehicle exterior such as an engine room of the vehicle, and is sometimes called a bulkhead.

(57) Specifically, the pedal device **1** includes a housing **5**, a housing bolt **22**, a reaction force generating mechanism **30**, a pedal **40**, a pad **402**, a shaft **41**, a rotating plate **43**, and a sensor unit **50**, as shown in FIGS. 2 to 7. Further, the pedal device **1** also includes a connecting rod **60**, a rod connecting screw **65**, a first stopper **70**, a second stopper **75** and a cover member **80**.

(58) The housing **5** includes a first housing **10**, a second housing **20** and a breathing hole **25**.

(59) The first housing **10** is made of resin in a box shape. The first housing **10** includes a top wall **11**, a left side wall **12**, a right side wall **13**, a front wall **14**, a housing space **15**, a housing opening **16** and a bearing hole **17**.

(60) The top wall **11** is a wall of the first housing **10** on the upper side of the vehicle, as shown in FIGS. 2 to 7. Further, the top wall **11** includes a housing hole **100** and a housing end **101**, as shown in FIG. 4. Most part of the reaction force generating mechanism **30** described later is inserted into the housing hole **100**. The housing end **101** defines the housing hole **100**. The left side wall **12** is a wall of the first housing **10** on the left side of the vehicle, as shown in FIG. 3.

(61) The right side wall **13** is, as shown in FIGS. 2 and 5, a wall of the first housing **10** on the right side of the vehicle. The right side wall **13** also includes a housing recess **130**.

(62) The housing recess **130** is recessed from an outer surface of the right side wall **13** in the left direction of the vehicle. Further, the housing recess **130** includes a housing recess bottom surface

305, a housing recess first side surface **301**, a housing recess second side surface **302**, a housing recess third side surface **303** and a housing recess space **306**.

(63) The housing recess bottom surface **305** is a surface facing the right side of the vehicle. Further, the housing recess bottom surface **305** is formed in an arc shape centered on the rotation axis CL of the pedal **40**, which will be described later.

(64) The housing recess first side surface **301** is connected to a vehicle rear side of the housing recess bottom surface **305**. Further, the housing recess first side surface **301** is formed in a shape of a side surface of an arc column centered on the rotation axis CL of the pedal **40**, which will be described later.

(65) The housing recess second side surface **302** is connected to a vehicle front side of the housing recess bottom surface **305**. Further, the housing recess second side surface **302** is formed in a shape of a side surface of an arc column centered on the rotation axis CL of the pedal **40**, which will be described later.

(66) The housing recess third side surface **303** is connected to a vehicle upper side of the following three surfaces, i.e., the housing recess bottom surface **305**, the housing recess first side surface **301**, and the housing recess second side surface **302**.

(67) The housing recess space **306** is a space formed by the housing recess bottom surface **305**, the housing recess first side surface **301**, the housing recess second side surface **302** and the housing recess third side surface **303**. Further, the housing recess space **306** is formed in an arc shape centered on the rotation axis CL of the pedal **40** described later, due to the shapes of the housing recess bottom surface **305**, the housing recess first side surface **301**, and the housing recess second side surface **302**.

(68) The front wall **14** is a wall of the first housing **10** on the front side of the vehicle, as shown in FIG. **4**. The housing space **15** is a space formed by the top wall **11**, the left side wall **12**, the right side wall **13** and the front wall **14**. The housing opening **16** is an open space formed by an end of the top wall **11**, an end of the left side wall **12**, an end of the right side wall **13** and an end of the front wall **14** in the housing space **15**. The bearing hole **17** is a cylindrical space extending in the vehicle left-right direction Dc.

(69) The second housing **20** is formed in a plate shape, and is connected to the respective ends of the top wall **11**, the left side wall **12**, the right side wall **13** and the front wall **14** of the first housing **10**. Also, for example, a hole (not shown) at the end of the top wall **11** of the first housing **10**, a hole (not shown) at the end of the left side wall **12**, a hole (not shown) at the end of the right side wall **13**, a hole (not shown) at the end of the front wall **14**, and holes on the second housing **20** corresponding thereto each have a bolt (not shown) inserted therinto. Thereby, the second housing **20** is fixed to the first housing **10** and closes the housing opening **16**. Also, the second housing **20** is made of metal. Furthermore, as described above, the first housing **10** is made of resin. Therefore, the Young's modulus of the second housing **20** is higher than that of the first housing **10**. Thus, the rigidity of the housing **5** is higher than when both the first housing **10** and the second housing **20** are made of resin. Furthermore, as shown in FIG. **3**, housing bolts **22** are inserted into (i) the holes of the second housing **20** and (ii) the holes of the floor **2** corresponding to the holes of the second housing **20**, thereby fixing the second housing **20** and the floor **2**. In such manner, the pedal device **1** is fixed to the floor **2**.

(70) The breathing hole **25** is a space formed at a position between the first housing **10** and the second housing **20**, as shown in FIG. **6**. Therefore, the breathing hole **25** communicates with the housing space **15** and a space outside the housing **5**. Further, the breathing hole **25** is formed at a certain position, e.g., either in an end of the front wall **14** of the first housing **10** or in the second housing **20** on a vehicle front side and on a vehicle upper side. Furthermore, the distance from the first housing **10** to the second housing **20** in the breathing hole **25** is approximately 0.05 mm to 1.0 mm.

(71) The reaction force generating mechanism **30** generates a reaction force against the driver's

stepping force applied to the pedal **40**. For example, the reaction force generating mechanism **30** includes a leaf spring **31**, a lower holder **35**, a large-diameter coil spring **33**, a small-diameter coil spring **34**, a spring seat **36** and an upper holder **37**, as shown in FIG. **4**.

(72) The leaf spring **31** is curved so as to form a convex curved surface toward the floor **2** when not receiving a load. Further, one end **311** of the leaf spring **31** is connected to a vehicle rear side of the second housing **20**. The lower holder **35** is connected to an other end **312** of the leaf spring **31**. The large-diameter coil spring **33** is connected to one side of the lower holder **35** opposite to the leaf spring **31**. The spring seat **36** includes a spring seat small-diameter portion **361** and a spring seat large-diameter portion **362**. The spring seat small-diameter portion **361** is formed in a cylindrical shape with a bottom. Further, the spring seat small-diameter portion **361** is arranged in a space inside the large-diameter coil spring **33**. The spring seat large-diameter portion **362** is connected to one side of the spring seat small-diameter portion **361** opposite to the bottom. Also, a diameter of the spring seat large-diameter portion **362** is larger than a diameter of the spring seat small-diameter portion **361**. Thereby, the spring seat large-diameter portion **362** and the large-diameter coil spring **33** are connected. Further, a small-diameter coil spring **34** is accommodated in the spring seat small-diameter portion **361**, and is connected to the bottom of the spring seat small-diameter portion **361**.

(73) The upper holder **37** has a holder small-diameter portion **371** and a holder large-diameter portion **372**. The holder small-diameter portion **371** is formed in a tubular shape. Further, the holder small-diameter portion **371** is arranged in a space inside the small-diameter coil spring **34**. Further, a pin is inserted into each of a hole of the leaf spring **31**, a hole of the lower holder **35**, a hole of the spring seat small-diameter portion **361**, and a hole of the holder small-diameter portion **371**, so that the leaf spring **31**, the lower holder **35**, the spring seat **36** and the upper holders **37** are connected to each other. In addition, the pin slides on an inner surface of the small-diameter portion **371** of the holder and also on an inner surface of the spring seat small-diameter portion **361**. The holder large-diameter portion **372** is connected to the holder small-diameter portion **371** on one side opposite to the spring seat **36**. Further, a diameter of the holder large-diameter portion **372** is larger than a diameter of the holder small-diameter portion **371**. Thereby, the holder large-diameter portion **372** and the small-diameter coil spring **34** are connected. Further, the holder large-diameter portion **372** includes a contact surface **373** that is in contact with the connecting rod **60**, which will be described later. The contact surface **373** is a surface of the holder large-diameter portion **372** opposite to the holder small-diameter portion **371**, and is flat.

(74) The pedal **40** is formed in a plate shape. The pedal **40** includes a pedal plate portion **401** and a rod fixing hole **403**.

(75) The pedal plate portion **401** is made of metal. The pedal plate portion **401** includes a pedal surface **404** and a pedal back surface **405**. The pedal surface **404** is one side facing the driver of the vehicle. The pedal back surface **405** is a surface of the pedal plate portion **401** opposite to the pedal surface **404**.

(76) The rod fixing hole **403** is a hole penetrating through the pedal surface **404** and the slope **407** of the pedal back surface **405**.

(77) The pad **402** is made of rubber or the like. Further, the pad **402** is connected to the vehicle upper side of the pedal surface **404**. Additionally, the pad **402** is stepped on by the driver of the vehicle. Further, the pad **402** covers a pedal surface **404** side of the rod fixing hole **403**. In such manner, the rod fixing hole **403** is not visible to the driver of the vehicle.

(78) The shaft **41** is made of metal, and has a cylindrical shape. Further, the shaft **41** is inserted into the bearing hole **17** of the first housing **10**, and is rotatably supported by the first housing **10**.

(79) The rotating plate **43** is formed in an L shape. Further, as shown in FIGS. **2** and **4**, the rotating plate **43** includes a back plate portion **44** and a side plate portion **45**. As shown in FIG. **4**, the back plate portion **44** is fixed to the pedal back surface **405** by inserting screws into a hole of the back plate portion **44** and into a corresponding hole of the pedal plate portion **401**. The side plate portion

45 is vertically connected to a vehicle right side of the back plate portion **44**. Further, the side plate portion **45** also includes a shaft hole **451** and a stopper hole **452**, as shown in FIG. 2.

(80) The shaft **41** is inserted into the shaft hole **451**. The pedal **40** are thus connected to the shaft **41** via the rotating plate **43**. Thereby, the pedal **40** rotates together with the shaft **41** and the rotating plate **43**, about the rotation axis CL, which is an axis of the shaft **41**.

(81) The stopper hole **452** is formed on a vehicle front side of the shaft hole **451**. Further, the stopper hole **452** communicates with the housing recess space **306**. Further, a part of the first stopper **70** described later is inserted into the stopper hole **452**.

(82) As shown in FIG. 3, the sensor unit **50** is arranged on one side of the shaft **41** opposite to the side plate portion **45**, that is, on the left side of the vehicle. Further, the sensor unit **50** includes a magnet, a yoke, a Hall element, or the like. (not shown). The sensor unit **50** detects a rotation angle of the pedal **40** by detecting a rotation angle of the shaft **41** using the magnet, the yoke, the Hall element, or the like. Further, the sensor unit **50** outputs a signal corresponding to the detected rotation angle of the pedal **40** to the first ECU **111** and the second ECU **112**. Note that the sensor unit **50** may include an MR element instead of the Hall element. MR is an abbreviation of Magneto Resistive. Further, the sensor unit **50** may be an inductive sensor that detects a rotation angle using a coil.

(83) The connecting rod **60** is made of metal, and has a rod shape. As shown in FIG. 4, the connecting rod **60** includes an arm portion **61** and a push portion **62**.

(84) The arm portion **61** includes an arm hole **611**, an arm recess **612**, and a cover member recess **613**. The arm hole **611** is a hole corresponding to the rod fixing hole **403**. The arm portion **61** is fixed to the pedal back surface **405** by inserting the rod connecting screw **65** into the arm hole **611** and the rod fixing hole **403**.

(85) The arm recess **612** is recessed in the axial direction of the connecting rod **60** from an end surface of the arm portion **61** on a push portion **62** side (to be described later). The arm recess **612** includes an arm recess side surface **614** and an arm recess bottom surface **615**. The arm recess side surface **614** is connected to an end surface of the arm portion **61** on a push portion **62** side, which will be described later. The arm recess bottom surface **615** is connected to the arm recess side surface **614**. Further, a space is formed by the arm recess side surface **614** and the arm recess bottom surface **615**.

(86) The cover member recess **613** is recessed from a side surface of the arm portion **61** in a direction orthogonal to the axial direction of the connecting rod **60**. Further, as shown in FIG. 7, the cover member recess **613** includes a cover member recess side surface **617** and a cover member recess bottom surface **616**. The cover member recess side surface **617** is connected to a side surface of the arm portion **61**. Further, the cover member recess side surface **617** faces the axial direction of the connecting rod **60**. The cover member recess bottom surface **616** is connected to the cover member recess side surface **617**. Further, the cover member recess bottom surface **616** faces in a direction orthogonal to the axial direction of the connecting rod **60**. Further, a space is formed by the cover member recess bottom surface **616** and the cover member recess side surface **617**.

(87) The push portion **62** includes a push contactor **621** and a push convex portion **622**. The push contactor **621** is in contact with the mechanism-side contact surface **373** of the upper holder **37** in the reaction force generating mechanism **30**. The push convex portion **622** protrudes in the axial direction of the connecting rod **60** from one side of the push contactor **621** opposite to the mechanism-side contact surface **373**. Also, the push convex portion **622** is inserted into a space formed by the arm recess side surface **614** and the arm recess bottom surface **615**. Thereby, the arm portion **61** and the push portion **62** are connected.

(88) As shown in FIGS. 2 and 5, the first stopper **70** is shaped like a rod extending in the vehicle left-right direction Dc. A part of the first stopper **70** is inserted into the stopper hole **452** of the rotating plate **43**. In such manner, the first stopper **70** rotates together with the rotating plate **43**, the pedal **40** and the shaft **41** about the rotation axis CL, which is the axis of the shaft **41**. Further, a

part of the first stopper **70** is inserted into the housing recess space **306**. The second stopper **75** protrudes from a vehicle upper side portion of the top wall **11** of the first housing **10** toward the pedal **40**.

(89) The cover member **80** is made of an elastic material such as rubber. Further, as shown in FIG. **4**, the cover member **80** is arranged on the front side of the vehicle with respect to the pedal back surface **405**. Further, the cover member **80** covers the housing hole **100**. Specifically, the cover member **80** has a first connecting portion **81**, a second connecting portion **82** and a covering portion **83**.

(90) The first connecting portion **81** is connected to the housing end portion **101** of the top wall **11** of the first housing **10**. Further, the first connecting portion **81** includes a first extension portion **811**, a second extension portion **812** and a third extension portion **813**, as shown in FIG. **7**.

(91) The first extension portion **811** is connected to an outer surface of the housing end portion **101**. Also, the first extension portion **811** extends in a direction orthogonal to the axial direction of the connecting rod **60**.

(92) The second extension portion **812** is connected to a side surface of the housing end portion **101** and the first extension portion **811**. Also, the second extension portion **812** extends from the first extension portion **811** in the axial direction of the connecting rod **60**. Further, the side surface of the housing end portion **101** is a surface connected to the outer surface and the inner surface of the housing end portion **101**.

(93) The third extension portion **813** is connected to the inner surface of the housing end portion **101** and the second extension portion **812**. Further, the third extension portion **813** extends in a direction orthogonal to the axial direction of the connecting rod **60** from one side of the second extension portion **812** opposite to the first extension portion **811**. Furthermore, here, the length of the third extension portion **813** in the direction orthogonal to the axial direction of the connecting rod **60** is the same as the length of the first extension portion **811**. Further, a space that is recessed in a direction orthogonal to the axial direction of the connecting rod **60** is formed by the first extension portion **811**, the second extension portion **812** and the third extension portion **813**. The housing end portion **101** is inserted into the above-described space. In such manner, the first connecting portion **81** hardly comes off from the housing end portion **101**.

(94) The second connecting portion **82** is connected to the arm portion **61** of the connecting rod **60**. Also, the second connecting portion **82** includes an insertion portion **821** and a protruding portion **822**. The insertion portion **821** is inserted into a space formed by the cover member recess bottom surface **616** and the cover member recess side surface **617** of the arm portion **61**. The protruding portion **822** protrudes from an entire circumference of the insertion portion **821** in a direction orthogonal to the axial direction of the connecting rod **60**.

(95) The covering portion **83** is formed in a bellows shape. Also, the covering portion **83** is connected to the first extension portion **811** and the second connecting portion **82** of the first connecting portion **81**. Thereby, the covering portion **83** covers a portion of the arm portion **61** of the connecting rod **60**, the push portion **62**, a portion of the upper holder **37** and the housing hole **100**.

(96) The brake-by-wire system **150** is configured as described above. Next, operation of the pedal device **1** will be described.

(97) When the driver of the vehicle does not step on the pad **402** of the pedal **40**, the large-diameter coil spring **33** and the small-diameter coil spring **34** of the reaction force generating mechanism **30** are compressed. At this time, restoring forces of the large-diameter coil spring **33** and the small-diameter coil spring **34** are transmitted to the pedal **40** via the connecting rod **60** connected to the upper holder **37** of the reaction force generating mechanism **30**. This restoring force causes the pedal **40** to rotate in a direction away from the first housing **10**. Also, at this time, the first stopper **70** connected to the pedal **40** via the rotating plate **43** contacts the housing recess third side surface **303** of the right side wall **13** of the first housing **10**. In such manner, a position of the pedal **40**

when the driver of the vehicle is not stepping on the pad **402** of the pedal **40** is fixed.

(98) When the driver of the vehicle steps on the pad **402** of the pedal **40**, the pedal **40** rotates together with the shaft **41** and the rotating plate **43** about the rotation axis CL, which is the axis of the shaft **41**. Such a rotation causes the pedal **40** to rotate in a direction toward the first housing **10**.

(99) At this time, the sensor unit **50** detects the rotation angle of the pedal **40** by detecting the rotation angle of the shaft **41**. Further, the sensor unit **50** outputs a signal corresponding to the detected rotation angle of the pedal **40** to the first ECU **111** and the second ECU **112**.

(100) At this time, the first ECU **111** rotates the motor **123** by supplying electric power to the motor **123**, for example. In such manner, the gear mechanism **125** is driven and the master piston **127** is moved. Therefore, the hydraulic pressure of the brake fluid supplied from the reservoir **124** to the master cylinder **126** increases. This increased hydraulic pressure is supplied to the second brake circuit **122**.

(101) Also, the second ECU **112** supplies electric power to an electromagnetic valve (not shown) of the second brake circuit **122**, for example. In such manner, a solenoid valve of the second brake circuit **122** opens. Therefore, the brake fluid supplied to the second brake circuit **122** is supplied to each of the wheel cylinders **131** to **134**. Thus, the brake pads attached to the wheel cylinders **131** to **134** rub against their corresponding brake discs. Therefore, each wheel is braked and the vehicle is decelerated. At this time, the second ECU **112** may perform ABS control, VSC control, collision avoidance control, regenerative cooperative control, or the like, based on signals from the sensor unit **50** and signals from other electronic control devices (not shown). ABS is an abbreviation of Anti-lock Braking System. VSC is an abbreviation of Vehicle Stability Control.

(102) Further, at this time, since the connecting rod **60** connected to the pedal **40** rotates together with the pedal **40** about the rotation axis CL, the connecting rod **60** passes through the housing hole **100**. Furthermore, since the second connecting portion **82** of the cover member **80** rotates together with the connecting rod **60**, the bellows-shaped covering portion **83** is compressed. Due to the restoring force of the covering portion **83** at this time, the covering portion **83** generates a reaction force against the driver's stepping force applied to the pedal **40**.

(103) Further, force from the pedal **40** is transmitted to the upper holder **37** by the push portion **62** of the connecting rod **60** coming into contact with the upper holder **37** of the reaction force generating mechanism **30**. In such manner, an inner surface of the holder small-diameter portion **371** of the upper holder **37** and the pin slide, and the small-diameter coil spring **34** is compressed by the upper holder **37**. Further, force from the pedal **40** is transmitted to the spring seat **36** via the small-diameter coil spring **34**. Also, the contact of the upper holder **37** with the spring seat **36** causes force from the pedal **40** to be transmitted to the spring seat **36** via the upper holder **37**. Therefore, an inner surface of the spring seat small-diameter portion **361** of the spring seat **36** and the pin slide, and the large-diameter coil spring **33** is compressed by the spring seat **36**. Further, force from the pedal **40** is transmitted to the leaf spring **31** via the large-diameter coil spring **33**. Therefore, the leaf spring **31** bends. Further, when an inner surface of the holder small-diameter portion **371** of the upper holder **37** and the pin slide and the holder large-diameter portion **372** of the upper holder **37** comes into contact with the pin, force from the pedal **40** is transmitted to the leaf spring **31** via the pin. In such manner, the leaf spring **31** bends. At this time, the reaction force generating mechanism **30** generates a reaction force against the driver's stepping force applied to the pedal **40** by the restoring force of the plate spring **31**, the large-diameter coil spring **33** and the small-diameter coil spring **34**. Therefore, even if the pedal **40** and the master cylinder **126** are not mechanically connected to each other, the pedal device **1** produces the corresponding reaction force as when the pedal **40** are connected to the master cylinder **126**, i.e., when the hydraulic reaction force is obtainable.

(104) Also, at this time, air in the housing space **15** is compressed by receiving force from the pedal **40**. Therefore, the air in the housing space **15** flows out to the space outside the housing **5** through the breathing hole **25**.

(105) Also, at this time, the first stopper **70** connected to the pedal **40** via the rotating plate **43** rotates with the axis of the shaft **41** as the rotation axis CL. Further, the housing recess space **306** into which a portion of the first stopper **70** is inserted is formed in an arc shape about the rotation axis CL of the pedal **40**. Therefore, the first stopper **70** moves within the housing recess space **306**. Also, the second stopper **75** contacts the pedal back surface **405**. Thus, at this time, even if the driver's stepping force applied to the pedals **40** is increased, the rotation of the pedals **40** is stopped. Further, at this time, the rotation angle of the pedal **40** is maximized.

(106) Further, when the driver of the vehicle stops stepping on the pad **402** of the pedal **40**, the reaction force of the reaction force generating mechanism **30** and the covering portion **83** cause the pedal **40** to rotate away from the first housing **10**. At this time, the first stopper **70** rotates together with the pedal **40**. In such manner, the first stopper **70** contacts the housing recess third side surface **303** of the right side wall **13** of the first housing **10** after moving within the housing recess space **306**. Therefore, the pedal **40** stops rotating. In such manner, the position of the pedal **40** returns to an initial position, i.e., to a position when the driver of the vehicle does not step on the pad **402** of the pedal **40**.

(107) Further, when the driver of the vehicle stops stepping on the pad **402** of the pedal **40**, the air in the housing space **15** expands due to the release of the stepping force applied by the driver of the vehicle. Therefore, the air outside the housing **5** flows into the housing space **15** through the breathing hole **25**.

(108) Thus, the pedal device **1** operates in the above-described manner. Next, how the pedal device **1** suppresses foreign matter from entering into the housing **5** will be described.

(109) The pedal device **1** includes the pedal **40**, the connecting rod **60**, the reaction force generating mechanism **30**, the housing **5** and the cover member **80**. The pedal **40** rotates about the rotation axis CL by being stepped on by the driver of the vehicle. The connecting rod **60** rotates together with the pedal **40** about the rotation axis CL by being connected to the pedal back surface **405**. The pedal back surface **405** is the surface of the pedal **40** opposite to the surface that is stepped on by the driver of the vehicle. The reaction force generating mechanism **30** is elastically deformed by receiving force from the pedal **40** through the connecting rod **60** when the pedal **40** is stepped on by the driver of the vehicle, and generates a reaction force against the stepping force of the driver of the vehicle applied to the pedal **40**. The housing **5** has the housing hole **100**, and accommodates the reaction force generating mechanism **30**. The housing hole **100** is a hole through which the connecting rod **60** passes when the pedal **40** is stepped on by the driver of the vehicle. The cover member **80** has the first connecting portion **81** and the covering portion **83**. The first connecting portion **81** is connected to the housing end portion **101**. The housing end portion **101** forms the housing hole **100** in the housing **5**. The covering portion **83** is arranged at a position between the first connecting portion **81** and the pedal back surface **405**, and is connected to the first connecting portion **81** to cover the housing hole **100**. Note that the driver of the vehicle corresponds to an operator. The first connecting portion **81** corresponds to a housing connection portion.

(110) Since the covering portion **83** is arranged at a position between the first connecting portion **81** and the pedal back surface **405**, the cover member **80** is not stepped on by the driver of the vehicle. Therefore, the cover member **80** will not be worn or damaged. Moreover, since the covering portion **83** covers the housing hole **100**, foreign matter cannot enter into the housing **5** through the housing hole **100** from the outside of the housing **5**. Thus, foreign matter is suppressed from entering into the housing **5**.

(111) Further, the pedal device **1** also has the following effects. [1-1] The covering portion **83** is formed in a bellows shape. Therefore, when the pedal **40** is stepped on by the driver of the vehicle, the covering portion **83** is elastically deformed by receiving force from the pedal **40** via the connecting rod **60**, thereby generating a reaction force against the stepping force of the driver applied to the pedal **40**.

(112) Thereby, when the connecting rod **60** or the reaction force generating mechanism **30** is

damaged, the covering portion **83** can generate a reaction force even when the reaction force of the reaction force generating mechanism **30** is not generated. Therefore, even if the connecting rod **60** or the reaction force generating mechanism **30** is damaged, the position of the pedal **40** can be returned to an initial position when the pad **402** of the pedal **40** is not stepped on by the driver of the vehicle. [1-2] The cover member **80** further includes the second connecting portion **82**. The second connecting portion **82** is connected to the covering portion **83** and the connecting rod **60**. The second connecting portion **82** corresponds to a rod connecting portion.

(113) In such manner, since the second connecting portion **82** is connected to the connecting rod **60**, the cover member **80** can be fixed to the connecting rod **60**. [1-3] The connecting rod **60** includes a cover member recess **613**. The second connecting portion **82** includes an inserting portion **821** and a protruding portion **822**. The insertion portion **821** is inserted into a space formed by the cover member recess bottom surface **616** and the cover member recess side surface **617** of the cover member recess **613**. The protruding portion **822** protrudes from the insertion portion **821** in a direction orthogonal to the axial direction of the connecting rod **60**. Note that the cover member recess **613** corresponds to a rod concave portion. The cover member recess side surface **617** corresponds to a rod side surface. The cover member recess bottom surface **616** corresponds to a rod bottom surface.

(114) In such manner, the second connecting portion **82** hardly comes off from the connecting rod **60**. Moreover, the second connecting portion **82** and the connecting rod **60** can be easily connected without providing an additional member. Therefore, the cost of the pedal device **1** is reducible by an amount of the cost of the additional member. [1-4] The first connecting portion **81** includes the first extension portion **811**, the second extension portion **812** and the third extension portion **813**. The first extension portion **811** extends in a direction orthogonal to the axial direction of the connecting rod **60**. The second extension portion **812** is connected to the first extension portion **811**, and extends in the axial direction of the connecting rod **60**. The third extension portion **813** is connected to the second extension portion **812** on one side opposite to the first extension portion **811**, and extends in a direction orthogonal to the axial direction of the connecting rod **60**. Also, the housing end portion **101** is inserted into the space formed by the first extension portion **811**, the second extension portion **812** and the third extension portion **813**. The direction orthogonal to the axial direction of the connecting rod **60** corresponds to one direction, or the extending direction of the first extension portion **811**, or the direction crossing the extending direction of the second extension portion **812**. Also, the axial direction of the connecting rod **60** corresponds to the direction crossing the extending direction of the first extension portion **811**.

(115) In such manner, the first connecting portion **81** hardly comes off from the housing end portion **101**. Further, the first connecting portion **81** and the housing end portion **101** can be easily connected without providing an additional member. Therefore, the cost of the pedal device **1** is reducible by an amount of the cost of the additional member. [1-5] The housing **5** has the first housing **10** and the second housing **20**. The first housing **10** includes the housing end portion **101**. The first housing **10** also forms the housing opening **16** and the housing space **15**. The housing opening **16** is provided on an opposite side of the housing hole **100**. The housing space **15** communicates with the housing hole **100** and the housing opening **16**. The second housing **20** closes the housing opening **16**. Further, the breathing hole **25**, which is a space between the end of the first housing **10** on a housing opening **16** side and the second housing **20**, is formed.

(116) Further, when the pedal **40** is stepped on by the driver of the vehicle and the air in the housing space **15** is compressed by receiving the force from the pedal **40**, the air in the housing space **15** passes through the breathing hole **25** and flows out to the outside of the housing **5**. Further, when the driver of the vehicle stops to step on the pedals **40** after stepping on the pedal **40**, the air in the housing space **15** expands as the force from the pedals **40** is released. At this time, air outside the housing **5** flows into the housing space **15** through the breathing holes **25**.

(117) The breathing hole **25** alleviates a volume change of the air in the housing space **15** caused by

the rotation of the pedal **40**. Further, the change in volume of the air in the housing space **15** caused by the temperature change in the housing space **15** is alleviated. Therefore, when the pedal **40** rotates, resistance to compression and expansion of the air in the housing space **15** is reducible, thereby the pedal **40** rotates smoothly. Furthermore, since the breathing hole **25** is formed at a position between the first housing **10** and the second housing **20**, there is no need to separately provide an additional hole and a dedicated hole drilling process is not required, which facilitates the manufacturing of the pedal device **1**.

Second Embodiment

(118) In the first embodiment described above, the second connecting portion **82** of the cover member **80** is connected to the connecting rod **60**. In contrast, in the second embodiment, as shown in FIG. **8**, a second connecting portion **82** is connected to a pedal back surface **405** by welding, adhesion, two-color molding, or the like. In such case, an arm portion **61** of a connecting rod **60** is not provided with a cover member recess **613**.

(119) The second embodiment is configured in the above-described manner. Also in the second embodiment, the same or corresponding effects as in the first embodiment can be obtained except for the above-described item [1-2]. Moreover, in the second embodiment, the following effects [2] can be obtained.

(120) [2] In the present embodiment, the second connecting portion **82** is connected to a covering portion **83** and to the pedal back surface **405**. The second connecting portion **82** corresponds to a pedal connecting portion.

(121) Thereby, a cover member **80** can be fixed to a pedal **40**. Further, even when it is difficult to fix the cover member **80** to the connecting rod **60** due to the connecting rod **60** being short, the cover member **80** can still be fixed.

Third Embodiment

(122) In the first embodiment described above, the second connecting portion **82** of the cover member **80** is connected to the connecting rod **60**. In contrast, in the third embodiment, as shown in FIG. **9**, a second connecting portion **82** is connected by welding, bonding, two-color molding, or the like, to a side surface of a holder large-diameter portion **372** of an upper holder **37** in a reaction force generating mechanism **30**. The side surface of the holder large-diameter portion **372** is a surface connected to a contact surface **373** of the upper holder **37** that is in contact with a connecting rod **60**.

(123) The third embodiment is configured in the above-described manner. Also in the third embodiment, the same or corresponding effects as in the first embodiment can be obtained except for the above-described item [1-2]. Moreover, in the third embodiment, the effects [3] described below can be also obtained.

(124) [3] The second connecting portion **82** is connected to a covering portion **83** and to the side surface of the holder large-diameter portion **372** of the upper holder **37** in the reaction force generating mechanism **30**. The second connecting portion **82** corresponds to a mechanism connecting portion.

(125) Thereby, a cover member **80** can be fixed to the reaction force generating mechanism **30**. Further, even when it is difficult to fix the cover member **80** to a connecting rod **60** or a pedal back surface **405**, the cover member **80** can still be fixed.

Fourth Embodiment

(126) In the fourth embodiment, a covering portion **83** of a cover member **80** has a sliding portion **835** as shown in FIG. **10**. The other configurations correspond to those of the third embodiment. A sliding portion **835** slides on a side surface of a connecting rod **60** when a pedal **40** rotates by a stepping of a driver of a vehicle.

(127) The fourth embodiment is configured in the above-described manner. The fourth embodiment achieves effects similar to the effects achieved by the first embodiment. Moreover, in the fourth embodiment, the following effects [4] can be obtained.

(128) [4] The sliding portion **835** of the covering portion **83** slides on the side surface of the connecting rod **60** when the driver of the vehicle steps on the pedal **40** to rotate.

(129) Here, when a pedal body is covered with a member such as rubber as in the automobile pedal disclosed in Patent Document 1, a footwear of the driver of the vehicle rubs, i.e., causes friction, against such a member. However, since such a friction varies depending on the footwear worn by the driver, it becomes difficult to design the life of the member that covers the pedal.

(130) On the other hand, since the covering portion **83** is arranged on a side of the pedal back surface **405**, it does not rub against the footwear of the driver of the vehicle. Further, if the sliding portion **835** of the covering portion **83** slides against the side surface of the connecting rod **60**, the covering portion **83** suppresses the intrusion of abrasion powder into a housing **5**, and an optimum material for such sliding can be selected. Therefore, it is possible to easily design the life of the covering portion **83** at a low cost.

(131) Further, friction between the sliding portion **835** of the covering portion **83** and the side surface of the connecting rod **60** causes a resistance corresponding to the rotation speed of the pedal **40** to occur in the pedal **40** via the connecting rod **60**. Therefore, the force acting on the pedal **40** becomes hysteresis. Further, when the driver of the vehicle slightly steps on the pedal **40**, the pedal **40** does not rotate due to this resistance force. Thus, the behavior of the vehicle becomes less prone to hunting.

Fifth Embodiment

(132) The fifth embodiment is different from the first embodiment in the form of a housing end portion **101** and the form of a first connecting portion **81** of a cover member **80**. The housing end portion **101** faces upward and rearward of the vehicle, as shown in FIG. **11**. Also, the size of a housing hole **100** of the fifth embodiment is larger than the size of the housing hole **100** of the first embodiment. In such case, a pedal device **1** does not have a second stopper **75**. However, when a pedal **40** rotates, for example, a first stopper **70** of the pedal device **1** contacts a surface of a second housing **20** on one side of a first housing **10**. Therefore, at this time, even if the driver's stepping force applied to the pedal **40** is increased, the rotation of the pedal **40** is stopped. Thus, at this time, the rotation angle of the pedal **40** is maximized.

(133) A first extension portion **811** of the first connecting portion **81** is connected to an outer surface of the housing end portion **101**. Further, the first extension portion **811** extends in the vehicle upward direction and the vehicle rearward direction. The vehicle upward direction and the vehicle rearward direction correspond to one direction and the extending direction of the first extension portion **811**.

(134) A second extension portion **812** is connected to a covering portion **83** of the cover member **80**, the side surface of the housing end portion **101** and the first extension portion **811**. Further, the second extension portion **812** extends in a direction crossing the extending direction of the first extension portion **811**.

(135) A third extension portion **813** is connected to the inner surface of the housing end portion **101** and the second extension portion **812**. The third extension portion **813** extends in the vehicle front direction and the vehicle lower direction from one side of the second extension portion **812** opposite to the first extension portion **811**. Further, a space is formed by the first extension portion **811**, the second extension portion **812** and the third extension portion **813**. The housing end portion **101** is inserted into the above-described space. In such manner, the first connecting portion **81** hardly comes off from the housing end portion **101**. The vehicle front direction and the vehicle lower direction correspond to directions crossing the extending direction of the second extension portion **812**.

(136) The fifth embodiment is configured in the above-described manner. The fifth embodiment achieves effects similar to the effects achieved by the first embodiment.

Sixth Embodiment

(137) In the sixth embodiment, the shapes of a housing hole **100** of a first housing **10** and a first

connecting portion **81** of a cover member **80** are different from those of the first embodiment. (138) As shown in FIGS. **12** and **13**, the housing hole **100** is circular in the direction orthogonal to the axial direction of a connecting rod **60**. Further, a second extension portion **812** of the first connecting portion **81** is formed in an annular shape in the direction orthogonal to the axial direction of the connecting rod **60**. Further, a plurality of third extension portions **813** of the first connecting portion **81** are formed. Each of the third extension portions **813** extends in a direction crossing the extending direction of the second extension portion **812** from a first extension portion **811**.

(139) The sixth embodiment is configured in the above-described manner. Also in the sixth embodiment, corresponding effects as in the first embodiment can be obtained.

Seventh Embodiment

(140) The seventh embodiment is different from the sixth embodiment in the form of a third extension portion **813** of a first connecting portion **81** of a cover member **80**.

(141) As shown in FIG. **14**, the third extension portion **813** is formed in an annular shape by protruding from the entire circumference of a second extension portion **812** in a direction orthogonal to the axial direction of a connecting rod **60**.

(142) The seventh embodiment is configured in the above-described manner. Also in the seventh embodiment, corresponding effects as in the sixth embodiment can be obtained.

Eighth Embodiment

(143) In the eighth embodiment, the shapes of a housing hole **100** of a first housing **10** and a first connecting portion **81** of a cover member **80** are different from those of the first embodiment.

(144) In the direction orthogonal to the axial direction of a connecting rod **60**, the housing hole **100** is formed in a square shape, as shown in FIG. **15**. Further, in the direction orthogonal to the axial direction of the connecting rod **60**, a second extension portion **812** of the first connecting portion **81** is formed in a square annular shape. Further, a third extension portion **813** of the first connecting portion **81** is formed in a square shape by protruding from the entire circumference of the second extension portion **812** in the direction orthogonal to the axial direction of the connecting rod **60**.

(145) The eighth embodiment is configured in the above-described manner. Also in the eighth embodiment, corresponding effects as in the first embodiment can be obtained. Further, the eighth embodiment also achieves the following effects (5).

(146) [5] The housing hole **100** is formed in a square shape in a cross section taken along a direction orthogonal to the axial direction of the connecting rod **60**. Note that the housing hole **100** is not limited to being formed in a square shape in a cross section taken along a direction orthogonal to the axial direction of the connecting rod **60**. The housing hole **100** may be formed in a polygonal shape in a cross section taken along a direction orthogonal to the axial direction of the connecting rod **60**.

(147) As a result, the length of the second extension portion **812** of the housing hole **100** and the first connecting portion **81** in one of the directions orthogonal to the axial direction of the connecting rod **60** can be made shorter than when the second extension portion **812** is formed in an annular shape. Therefore, the size of the first housing **10** and the cover member **80** is reducible, thereby the size of a pedal device **1** is reducible.

Ninth Embodiment

(148) The ninth embodiment is different from the first embodiment in the form of a connecting rod **60** and the form of a second connecting portion **82** of a cover member **80**.

(149) In the direction orthogonal to the axial direction of the connecting rod **60**, a housing hole **100** is formed in an elliptical shape, as shown in FIG. **16**. Further, in the direction orthogonal to the axial direction of the connecting rod **60**, a second extension portion **812** of a first connecting portion **81** is formed in an elliptical ring shape. Further, a third extension portion **813** of the first connecting portion **81** is formed in an elliptical shape, by protruding from the entire circumference of the second extension portion **812** in a direction orthogonal to the axial direction of the

connecting rod **60**.

(150) The ninth embodiment is configured in the above-described manner. Also in the ninth embodiment, corresponding effects as in the first embodiment can be obtained. Further, the ninth embodiment also achieves the following effects [6].

(151) [6] The housing hole **100** is formed in an elliptical shape in a cross section taken along a direction orthogonal to the axial direction of the connecting rod **60**.

(152) As a result, the length of the second extension portion **812** of the housing hole **100** and the first connecting portion **81** in one of the directions orthogonal to the axial direction of the connecting rod **60** can be made shorter than when the second extension portion **812** is formed in an annular shape. Therefore, the size of the first housing **10** and the cover member **80** is reducible, thereby the size of a pedal device **1** is reducible.

Tenth Embodiment

(153) The tenth embodiment is different from the first embodiment in the form of a first extension portion **811** and a third extension portion **813** in a first connecting portion **81** of a cover member **80**.

(154) As shown in FIG. **17**, the length of the third extension portion **813** is shorter than the length of the first extension portion **811** in the direction orthogonal to the axial direction of a connecting rod **60**. The third extension portion **813** also includes a third extension slope **814**. The third extension slope **814** is formed on one side of the third extension portion **813** opposite to the portion on a side of a housing end portion **101**.

(155) The tenth embodiment is configured in the above-described manner. Also in the tenth embodiment, corresponding effects as in the first embodiment can be obtained. Further, the tenth embodiment also achieves the following effects (7).

(156) [7] The length of the third extension portion **813** is made shorter than the length of the first extension portion **811**. The third extension portion **813** also includes the third extension slope **814**. The third extension slope **814** is formed on the opposite side of the portion of the third extension portion **813** that is connected to the housing end portion **101**, and is inclined toward the extending direction of the third extension portion **813**. Thereby, the third extension slope **814** guides the housing end portion **101** into the space formed by the first extension portion **811**, the second extension portion **812** and the third extension portion **813**. Further, the third extension slope **814** corresponds to a housing slope.

(157) Since the length of the third extension portion **813** is shorter than the length of the first extension portion **811**, the housing end portion **101** is easily insertable from a side of the third extension portion **813** into the space formed by the first extension portion **811**, the second extension portion **812** and the third extension portion **813**. Further, when the housing end portion **101** is inserted into such space, the housing end portion **101** is guided into such space by the third extension slope **814**, thereby the housing end portion **101** can be easily inserted into such space.

Eleventh Embodiment

(158) The eleventh embodiment is different from the first embodiment in the form of a first extension portion **811** and a third extension portion **813** of a first connecting portion **81** of a cover member **80**.

(159) As shown in FIG. **18**, the length of the first extension portion **811** is shorter than the length of the third extension portion **813** in the direction orthogonal to the axial direction of a connecting rod **60**. The first extension portion **811** also includes a first extension slope **815**. The first extension slope **815** is formed on one side of the first extension portion **811** opposite to the outer surface of a housing end portion **101**.

(160) The eleventh embodiment is configured in the above-described manner. The eleventh embodiment also achieves corresponding effects as the first embodiment. Further, the eleventh embodiment also achieves the following effects (8).

(161) [8] The length of the first extension portion **811** is shorter than the length of the third

extension portion **813**. The first extension portion **811** also includes the first extension slope **815**. The first extension slope **815** is formed on one side opposite to the outer surface of the housing end portion **101** and is inclined with respect to the extending direction of the first extension portion **811**. Thereby, the first extension slope **815** guides the housing end portion **101** into the space formed by the first extension portion **811**, a second extension portion **812** and the third extension portion **813**. Note that, in such case, the first extension portion **811** corresponds to the third extension portion **813**. The first extension slope **815** corresponds to a third extension slope **814** and also to a housing slope.

(162) Since the length of the first extension portion **811** is shorter than the length of the third extension portion **813**, the housing end portion **101** is more easily insertable from a side of the first extension portion **811** into the space formed by the first extension portion **811**, the second extension portion **812** and the third extension portion **813**. Further, when the housing end portion **101** is inserted into such space, the housing end portion **101** is guided into such space by the first extension slope **815**, thereby the housing end portion **101** can be easily insertable into such space.

Twelfth Embodiment

(163) The twelfth embodiment is different from the first embodiment in the form of a housing end portion **101** of a first housing **10** of a housing **5** and the form of a first connecting portion **81** of a cover member **80**.

(164) The housing end portion **101** includes an end recess **105**, as shown in FIG. **19**. The end recess **105** is recessed from a side surface of the housing end portion **101** in a direction orthogonal to the axial direction of a connecting rod **60**. The end recess **105** also includes an end recess side surface **107** and an end recess bottom surface **106**. Further, the side surface of the housing end portion **101** is a surface connected to the outer surface and the inner surface of the housing end portion **101**.

(165) The end recess side surface **107** is connected to the side surface of the housing end portion **101**. The end recess bottom surface **106** is connected to the end recess side surface **107**. The first connecting portion **81** is inserted into the space formed by the recess end side surface **107** and the end recess bottom surface **106**. Thereby, the first connecting portion **81** and the housing end portion **101** are connected.

(166) The twelfth embodiment is configured in the above-described manner. The twelfth embodiment also achieves corresponding effects as the first embodiment. Further, the twelfth embodiment also achieves the following effects [9].

(167) [9] The housing end portion **101** includes the end recess **105** that is recessed from the surrounding. The end recess **105** includes the end recess side surface **107** and the end recess bottom surface **106**. The first connecting portion **81** is inserted into the space formed by the end recess side surface **107** and the end recess bottom surface **106**. Note that the end recess side surface **107** corresponds to an end side surface. The end recess bottom surface **106** corresponds to an end bottom surface.

(168) In such manner, the first connecting portion **81** hardly comes off from the housing end portion **101**. Further, the first connecting portion **81** and the housing end portion **101** can be easily connected without providing an additional member. Therefore, the cost of a pedal device **1** is reducible by an amount of the cost of the additional member.

Thirteenth Embodiment

(169) In the thirteenth embodiment, an arm portion **61** of a connecting rod **60** does not have an arm recess **612**. Also, the form of a cover member **80** is different from that of the first embodiment.

(170) A first connecting portion **81** of the cover member **80** is inserted into a housing hole **100** and connected to the side surface of a housing end portion **101**, as shown in FIG. **20**. Also, a second connecting portion **82** of the cover member **80** is connected to the side surface of the arm portion **61** of the connecting rod **60**. Further, a covering portion **83** of the cover member **80** is connected to the first connecting portion **81** and the second connecting portion **82**. Further, the covering portion **83** is formed in a bellows shape, and the covering portion **83** is elastically deformed by being

compressed.

(171) The thirteenth embodiment is configured in the above-described manner. The thirteenth embodiment also achieves corresponding effects as the first embodiment. Further, the thirteenth embodiment also achieves the following effects [10].

(172) [10] The covering portion **83** is formed in a bellows shape. Moreover, the covering portion **83** is elastically deformed by being compressed, and thus generates a restoring force acting in the direction of pressing the first connecting portion **81** against the housing end portion **101**. Further, the covering portion **83** is elastically deformed by being compressed, and thus generates a restoring force acting in the direction of pressing the second connecting portion **82** against the connecting rod **60**.

(173) Such a restoring force maintains the connection between the first connecting portion **81** and the housing end portion **101**. Also, such a restoring force maintains the connection between the second connecting portion **82** and the connecting rod **60**.

Fourteenth Embodiment

(174) The fourteenth embodiment is different from the thirteenth embodiment in the form of a first connecting portion **81** and a second connecting portion **82** of a cover member **80**.

(175) The first connecting portion **81** includes a first high-strength portion **841** and a first low-strength portion **851**, as shown in FIG. 21. The first high-strength portion **841** is made of metal. Also, the first high-strength portion **841** is formed in an annular shape. Note that the first high-strength portion **841** may be made of resin.

(176) The first low-strength portion **851** is made of rubber or the like. Therefore, the Young's modulus of the first low-strength portion **851** is lower than that of the first high-strength portion **841**. Also, the first low-strength portion **851** is formed in an annular shape. Further, the first low-strength portion **851** surrounds the first high-strength portion **841**. The first low-strength portion **851** and a housing end portion **101** are connected by inserting the first low-strength portion **851** together with the first high-strength portion **841** into a housing hole **100**.

(177) The second connecting portion **82** includes a second high-strength portion **842** and a second low-strength portion **852**. The second high-strength portion **842** is made of metal. Also, the second high-strength portion **842** is formed in an annular shape. Note that the second high-strength portion **842** may be made of resin.

(178) The second low-strength portion **852** is made of rubber or the like. Therefore, the Young's modulus of the second low-strength portion **852** is lower than that of the second high-strength portion **842**. Also, the second low-strength portion **852** is formed in an annular shape. Further, the second low-strength portion **852** surrounds the second high-strength portion **842**. Also, an arm portion **61** of a connecting rod **60** is inserted into the space inside the second low-strength portion **852**. The Young's modulus of the first high-strength portion **841**, the first low-strength portion **851**, the second high-strength portion **842**, and the second low-strength portion **852** are measured by a tensile test or the like.

(179) The fourteenth embodiment is configured in the above-described manner. The fourteenth embodiment also has corresponding effects as the thirteenth embodiment. Further, the fourteenth embodiment also achieves the following effects [11-1], [11-2].

(180) [11-1] The first connecting portion **81** includes the first high-strength portion **841** and the first low-strength portion **851**. The first high-strength portion **841** is surrounded by the first low-strength portion **851**, and has a higher Young's modulus than the first low-strength portion **851**. The first low-strength portion **851** is connected to the housing end portion **101** by being inserted into the housing hole **100**.

(181) The strength of the first connecting portion **81** is improved by the first high-strength portion **841**. Therefore, the first low-strength portion **851** can be press-fitted into the housing hole **100**. Thereby, the force for maintaining the connection between the first connecting portion **81** and the housing end portion **101** is improved.

(182) [11-2] The second connecting portion **82** includes the second high-strength portion **842** and the second low-strength portion **852**. The second high-strength portion **842** is surrounded by the second low-strength portion **852**, and has a higher Young's modulus than the second low-strength portion **852**. The second low-strength portion **852** is formed in an annular shape. Also, the second low-strength portion **852** is connected to the connecting rod **60** by inserting the connecting rod **60** into the second low-strength portion **852**.

(183) The strength of the second connecting portion **82** is improved by the second high-strength portion **842**. Therefore, the connecting rod **60** can be press-fitted into the second low-strength portion **852**. Thereby, the force for maintaining the connection between the second connecting portion **82** and the connecting rod **60** is improved.

Fifteenth Embodiment

(184) The fifteenth embodiment is different from the thirteenth embodiment in the form of a housing end portion **101**.

(185) The housing end portion **101** includes a housing extension portion **210**, as shown in FIG. 22. The housing extension portion **210** extends in the axial direction of a connecting rod **60**. Also, a first connecting portion **81** of a cover member **80** is connected to a surface of the housing extension portion **210** in a direction orthogonal to the axial direction of the connecting rod **60**. In such case, a covering portion **83** is elastically deformed by being compressed, thereby generating a restoring force acting in the direction of pressing the first connecting portion **81** against the housing extension portion **210**.

(186) The fifteenth embodiment is configured in the above-described manner. The fifteenth embodiment also achieves the same or corresponding effects as the thirteenth embodiment.

Sixteenth Embodiment

(187) The sixteenth embodiment is different from the twelfth embodiment in the shape of an end recess **105** of a housing end portion **101**.

(188) The end recess **105** is recessed in the axial direction of a connecting rod **60** from the outer surface of the housing end portion **101**, as shown in FIG. 23. Also, a first connecting portion **81** of a cover member **80** is inserted into the space formed by an end recess side surface **107** and an end recess bottom surface **106** of the end recess **105**. Thereby, the first connecting portion **81** and the housing end portion **101** are connected.

(189) Also, the first connecting portion **81** is formed of an elastic member. Further, the first connecting portion **81** is compressed. In such manner, the first connecting portion **81** generates a restoring force acting in the direction orthogonal to the axial direction of the connecting rod **60**. Since such a restoring force acts in the direction of pressing the first connecting portion **81** against the end recess side surface **107**, the connection between the first connecting portion **81** and the housing end portion **101** is maintained.

(190) The sixteenth embodiment is configured in the above-described manner. The sixteenth embodiment also achieves corresponding effects as the twelfth embodiment.

Seventeenth Embodiment

(191) The seventeenth embodiment is different from the first embodiment in the form of a housing end portion **101**, the form of a connecting rod **60**, and the forms of a first connecting portion **81** and a second connecting portion **82** of a cover member **80**. Moreover, a pedal device **1** further includes a first fixing member **91** and a second fixing member **92**.

(192) The housing end portion **101** includes a fixing housing hole **201**, as shown in FIG. 24. The fixing housing hole **201** is a space extending in the axial direction of the connecting rod **60**. An arm portion **61** of the connecting rod **60** does not have a cover member recess **613**, but has a fixing rod hole **620**. The fixing rod hole **620** is a space extending in a direction orthogonal to the axial direction of the connecting rod **60**. The first connecting portion **81** has a first connecting portion hole **861** corresponding to the fixing housing hole **201**. The second connecting portion **82** has a second connecting portion hole **862** corresponding to the fixing rod hole **620**.

(193) The first fixing member **91** and the second fixing member **92** are screws, pins, snap-fits, or the like. A portion of the first fixing member **91** is inserted into the fixing housing hole **201** and the first connecting portion hole **861**. Thereby, the first connecting portion **81** and the housing end portion **101** are connected. Further, a portion of the second fixing member **92** is inserted into the fixing rod hole **620** and the second connecting portion hole **862**. Therefore, the second connecting portion **82** and the arm portion **61** of the connecting rod **60** are connected.

(194) The seventeenth embodiment is configured in the above-described manner. The seventeenth embodiment also achieves corresponding effects as the first embodiment except for the above-described items [1-3] and [1-4]. Further, the seventeenth embodiment also achieves the following effects [12-1], [12-2].

(195) [12-1] The fixing housing hole **201** is formed in the housing end portion **101**. The first connecting portion hole **861** corresponding to the fixing housing hole **201** is formed in the first connecting portion **81** of the cover member **80**. The pedal device **1** further includes the first fixing member **91**. A portion of the first fixing member **91** is inserted into the fixing housing hole **201** and the first connecting portion hole **861**, thereby connecting the first connecting portion **81** and the housing end portion **101**.

(196) This makes it easy to fix the first connecting portion **81** and the housing end portion **101**. Also, the force for maintaining the connection between the first connecting portion **81** and the housing end portion **101** is improved.

(197) [12-2] The connecting rod **60** is formed with the fixing rod hole **620** which is a space extending in a direction orthogonal to the axial direction of the connecting rod **60**. The second connecting portion hole **862** corresponding to the fixing rod hole **620** is formed in the second connecting portion **82** of the cover member **80**. The pedal device **1** further includes the second fixing member **92**. A portion of the second fixing member **92** is inserted into the fixing rod hole **620** and the second connecting portion hole **862**, thereby connecting the second connecting portion **82** and the connecting rod **60**.

(198) This makes it easy to fix the second connecting portion **82** and the connecting rod **60**. Also, the force for maintaining the connection between the second connecting portion **82** and the connecting rod **60** is improved.

Eighteenth Embodiment

(199) The eighteenth embodiment is different from the first embodiment in (i) the connection between a housing end portion **101** and a first connecting portion **81** of a cover member **80** and (ii) the connection between a connecting rod **60** and a second connecting portion **82** of the cover member **80**.

(200) As shown in FIG. 25, the first connecting portion **81** is connected to the outer surface of the housing end portion **101** by welding, adhesion, two-color molding, or the like. The second connecting portion **82** is connected to the side surface of an arm portion **61** of the connecting rod **60** by welding, adhesion, two-color molding, or the like. Note that the arm portion **61** does not have a cover member recess **613**.

(201) The eighteenth embodiment is configured in the above-described manner. The eighteenth embodiment also achieves corresponding effects as the first embodiment except for the above-described items [1-3] and [1-4]. Further, the eighteenth embodiment also achieves the following effects [13-1], [13-2].

(202) [13-1] The first connecting portion **81** is connected to the housing end portion **101** by welding, adhesion or two-color molding. This makes it easy to fix the first connecting portion **81** and the housing end portion **101**.

(203) [13-2] The second connecting portion **82** is connected to the connecting rod **60** by welding, adhesion, two-color molding, or the like. This makes it easy to fix the second connecting portion **82** and the connecting rod **60**.

Nineteenth Embodiment

(204) The nineteenth embodiment is different from the first embodiment in (i) the connection between a housing end portion **101** and a first connecting portion **81** of a cover member **80** and (ii) the connection between a connecting rod **60** and a second connecting portion **82** of a cover member **80**.

(205) The housing end portion **101** includes a housing extension portion **210**, as shown in FIG. **26**. The housing extension portion **210** extends in the axial direction of the connecting rod **60**. The first connecting portion **81** of the cover member **80** is connected in the direction orthogonal to the axial direction of the connecting rod **60** to the outer surface of the housing extension portion **210** facing in the direction orthogonal to the axial direction of the connecting rod **60**.

(206) Moreover, a pedal device **1** further includes a first fixing member **91** and a second fixing member **92**. The first fixing member **91** and the second fixing member **92** are binding bands, retaining rings, or the like. Therefore, the first fixing member **91** and the second fixing member **92** have an annular shape. Note that the first fixing member **91** and the second fixing member **92** are not limited to have the annular shape, and may be, for example, arc-shaped, elliptical arc-shaped, U-shaped, L-shaped, or the like.

(207) Also, the first fixing member **91** and the housing extension portion **210** sandwich the first connecting portion **81**. Further, the second fixing member **92** and the side surface of the arm portion **61** of the connecting rod **60** sandwich the second connecting portion **82**.

(208) The nineteenth embodiment is configured in the above-described manner. The nineteenth embodiment also achieves corresponding effects as the first embodiment except for the above-described items [1-3] and [1-4]. Further, the nineteenth embodiment also achieves the following effects [14-1], [14-2].

(209) [14-1] The housing end portion **101** includes the housing extension portion **210**. The housing extension portion **210** extends in the axial direction of the connecting rod **60**. The housing extension portion **210** and the first connecting portion **81** are connected by sandwiching of the first connecting portion **81** with the first fixing member **91** and the housing extension portion **210**. This makes it easy to fix the first connecting portion **81** and the housing end portion **101**. Also, the force for maintaining the connection between the first connecting portion **81** and the housing end portion **101** is improved.

(210) [14-2] The connecting rod **60** and the second connecting portion **82** are connected by sandwiching of the second connecting portion **82** with the second fixing member **92** and the side surface of the connecting rod **60**. This makes it easy to fix the second connecting portion **82** and the connecting rod **60**. Further, the force for maintaining the connection between the second connecting portion **82** and the connecting rod **60** is improved.

Twentieth Embodiment

(211) The twentieth embodiment is different from the first embodiment in the form of a covering portion **83** of a cover member **80**.

(212) The covering portion **83** has a first covering portion **871** and a second covering portion **872**, as shown in FIG. **27**. Therefore, the covering portion **83** is divided into two. Also, the first covering portion **871** is connected to the second covering portion **872** in a direction orthogonal to the axial direction of a connecting rod **60** by welding, adhesion, two-color molding, or the like.

(213) The twentieth embodiment is configured in the above-described manner. The twentieth embodiment also achieves corresponding effects as the first embodiment, except for the above-described item [1-3]. Further, the twentieth embodiment also achieves the following effects [15].

(214) [15] The covering portion **83** has the first covering portion **871** and the second covering portion **872**. The first covering portion **871** and the second covering portion **872** are connected to each other in a direction orthogonal to the axial direction of the connecting rod **60**, and cover the housing hole **100**.

(215) Thus, by dividing the covering portion **83**, it is easy to perform fixing between the first connecting portion **81** and a housing end portion **101**, since the connection between the first

connecting portion **81** and the housing end portion **101** can be separately performed. Further, since the connection between the second connecting portion **82** and the connecting rod **60** can also be separately performed, the second connecting portion **82** and the connecting rod **60** can be easily fixed.

Twenty-First Embodiment

(216) The twenty-first embodiment is different from the first embodiment in the form of a covering portion **83** of a cover member **80**. The covering portion **83** has a cut **873** as shown in FIG. **28**. Also, a portion of the covering portion **83** at the cut **873** is connected by welding, adhesion, two-color molding, or the like.

(217) The twenty-first embodiment is configured in the above-described manner. Also in the twenty-first embodiment, corresponding effects as in the twentieth embodiment can be obtained.

Twenty-Second Embodiment

(218) In the twenty-second embodiment, the shape of a breathing hole **25** is different from that in the first embodiment.

(219) The breathing hole **25** is formed between the inner surface of a front wall **14** of a first housing **10** and the side surface of a second housing **20**, as shown in FIG. **29**. The side surface of the second housing **20** is a surface connected to the outer surface and the inner surface of the second housing **20**.

(220) The twenty-second embodiment is configured in the above-described manner. Also in the twenty-second embodiment, corresponding effects as in the first embodiment can be obtained.

Twenty-Third Embodiment

(221) In the twenty-third embodiment, the shapes of a first housing **10**, a second housing **20** and a breathing hole **25** are different from those of the first embodiment.

(222) The first housing **10** has a protrusion **141** as shown in FIG. **30**. The protrusion **141** protrudes from the side surface of a front wall **14** of the first housing **10** toward the second housing **20**. Further, a portion of the protrusion **141** is inserted into a recessed space formed in the second housing **20**. Thereby, the breathing hole **25** is formed between the second housing **20** and the protrusion **141**. Further, the protrusion **141** and the recessed space formed in the second housing **20** provide a labyrinthine shape for the breathing hole **25**. Further, the side surface of the front wall **14** is a surface connected to the outer surface and the inner surface of the front wall **14**.

(223) The twenty-third embodiment is configured in the above-described manner. The twenty-third embodiment also achieves corresponding effects as the first embodiment. Further, the twenty-third embodiment also achieves the following effects [16].

(224) [16] A portion of the protrusion **141** of the first housing **10** is inserted into the recessed space formed in the second housing **20**, thereby providing a labyrinthine shape for the breathing hole **25**.

(225) In such manner, when foreign matter passes through the breathing hole **25** from the outside of a housing **5**, the movement of foreign matter is prevented by the protrusion **141**, thereby foreign matter is suppressed from passing through the breathing hole **25** from the outside of the housing **5** and intruding into a housing space **15**. Also, when foreign matter passes through the breathing hole **25** from the outside of the housing **5**, it accumulates in the recessed space of the second housing **20**. Therefore, foreign matter is suppressed from intruding into the housing space **15** from the outside of the housing **5** through the breathing hole **25**.

Twenty-Fourth Embodiment

(226) The twenty-fourth embodiment is different from the first embodiment in the form of a breathing hole **25**.

(227) As shown in FIG. **31**, the breathing hole **25** is formed between (a) the left side of a front wall **14** of a first housing **10**, the left side of a left side wall **12** and the left side of a first housing **10** on a rear side of the vehicle and (b) a second housing **20**. Further, since a hole is formed at the end of the left side wall **12** of the first housing **10**, a portion of the breathing hole **25** on the left side of the vehicle is branched. Therefore, such breathing hole **25** is larger than the one in the first

embodiment. In such manner, it becomes easy for the air to flow in and out through the breathing hole **25**, thereby the change in volume of the air in a housing space **15** caused by the rotation of a pedal **40** and the temperature change in the housing space **15** can easily be alleviated. Thus, when the pedal **40** rotates, resistance to compression and expansion of the air in the housing space **15** is reduced, for a smooth rotation of the pedal **40**. Further, in FIG. **31**, the range of the breathing hole **25** is indicated by a double-dot chain line.

(228) The twenty-fourth embodiment is configured in the above-described manner. The twenty-fourth embodiment also achieves corresponding effects as the first embodiment.

Twenty-Fifth Embodiment

(229) The twenty-fifth embodiment is different from the twenty-fourth embodiment in the form of a first housing **10**.

(230) The first housing **10** has an accumulation hole **215** communicating with a breathing hole **25**, as shown in FIG. **32**. Further, in FIG. **32**, the range of the breathing hole **25** is indicated by a double-dot chain line. Also, a second housing **20** may have the accumulation hole **215**.

(231) The twenty-fifth embodiment is configured in the above-described manner. Also in the twenty-fifth embodiment, corresponding effects as the first embodiment can be obtained. Further, the twenty-fifth embodiment also achieves the following effects [17].

(232) [17] A housing **5** has the accumulation hole **215** communicating with the breathing hole **25**. Foreign matter intruding from the outside of the housing **5** is accumulated in the accumulation hole **215**.

(233) In such manner, when foreign matter passes through the breathing hole **25** from the outside of the housing **5**, the foreign matter accumulates in the accumulation hole **215**, thereby suppressing the foreign matter from intruding into a housing space **15** through the breathing hole **25** from the outside of the housing **5**.

Twenty-Sixth Embodiment

(234) The twenty-sixth embodiment is different from the twenty-fourth embodiment in the shape of a breathing hole **25**.

(235) The breathing hole **25** is formed substantially in the entire circumference between a first housing **10** and a second housing **20**, as shown in FIG. **33**. Further, in FIG. **33**, the range of the breathing hole **25** is indicated by the double-dot chain line.

(236) The twenty-sixth embodiment is configured in the above-described manner. The twenty-sixth embodiment also achieves corresponding effects as the twenty-fourth embodiment.

Twenty-Seventh Embodiment

(237) In the twenty-seventh embodiment, a first housing **10** has a filter hole **220**. A pedal device **1** also includes a filter **225**. The other configurations are similar to those of the first embodiment.

(238) The filter hole **220** is formed in a front wall **14** of the first housing **10**, as shown in FIG. **34**. Also, the filter hole **220** allows communication between the outside of the first housing **10** and a housing space **15**. Note that the filter hole **220** is not limited to being formed in the front wall **14** of the first housing **10**, and may also be formed in a left side wall **12**, a right side wall **13**, a rear side wall of the vehicle, and a second housing **20**.

(239) The filter **225** is inserted into the filter hole **220**. The filter **225** is a vent filter. Therefore, the filter **225** does not allow water flowing from the outside of the first housing **10** into the housing space **15** through the filter hole **220**, but allows air to pass therethrough.

(240) The twenty-seventh embodiment is configured in the above-described manner. Also in the twenty-seventh embodiment, corresponding effects as the first embodiment can be obtained. The twenty-seventh embodiment also achieves the following effects [18].

(241) [18] A housing **5** has the filter hole **220** formed thereon. The filter hole **220** allows communication between the outside of the housing **5** and the housing space **15**. Moreover, the pedal device **1** further includes the filter **225**. The filter **225** allows air to pass therethrough while preventing water to flow from the outside of the housing **5** into the housing space **15** through the

filter hole **220**. Note that water corresponds to foreign matter.

(242) Further, when a pedal **40** is stepped on by the driver of the vehicle, the air in the housing space **15** is compressed by receiving force from the pedal **40**, and the air in the housing space **15** passes through the filter hole **220** and the filter **225**, and flows out of the housing **5**. Further, when the driver of the vehicle stops to step on the pedals **40** after stepping on the pedal **40**, the air in the housing space **15** expands as the force from the pedal **40** is released. At this time, the air outside the housing **5** passes through the filter hole **220** and the filter **225** and flows into the housing space **15**.

(243) The filter hole **220** and the filter **225** alleviate the volume change of the air in the housing space **15** caused by the rotation of the pedal **40**. Further, the change in volume of the air in the housing space **15** caused by the temperature change in the housing space **15** is alleviated.

Therefore, when the pedal **40** rotates, resistance to compression and expansion of the air in the housing space **15** is reduced, thereby the pedal **40** rotates smoothly. Further, the filter **225** suppresses foreign matter from intruding into the housing space **15** from the outside of the housing **5** through the filter hole **220**.

Twenty-Eighth Embodiment

(244) The twenty-eighth embodiment is different from the first embodiment in the form of a connecting rod **60** and a cover member **80**.

(245) An arm portion **61** of the connecting rod **60** does not have an arm recess **612**, but has an arm protrusion **632** as shown in FIG. **35**. The arm protrusion **632** protrudes from the side surface of the arm portion **61** in a direction orthogonal to the axial direction of the connecting rod **60**.

(246) A second connecting portion **82** of the cover member **80** has a bottom portion **883**, a first side portion **881** and a second side portion **882**. The bottom portion **883** extends in the axial direction of the connecting rod **60**. The first side portion **881** extends from the bottom portion **883** in a direction orthogonal to the axial direction of the connecting rod **60**. The second side portion **882** extends in a direction orthogonal to the axial direction of the connecting rod **60** from one side of the bottom portion **883** opposite to the first side portion **881**. Also, the arm protrusion **632** is inserted into the space formed by the bottom portion **883**, the first side portion **881** and the second side portion **882**. Thereby, the second connecting portion **82** is connected to the arm portion **61**.

(247) Also, the length of the second side portion **882** is shorter than the length of the first side portion **881** in the direction orthogonal to the axial direction of the connecting rod **60**. Additionally, the second side portion **882** includes a second side slope **884**. The second side slope **884** is formed on one side of the second side portion **882** opposite to a portion on a side of the arm protrusion **632**. Note that the length of the first side portion **881** may be shorter than the length of the second side portion **882** in the direction orthogonal to the axial direction of the connecting rod **60**. Also, the first side portion **881** may include a slope similar to the second side slope **884**.

(248) The twenty-eighth embodiment is configured in the above-described manner. The twenty-eighth embodiment also achieves corresponding effects as the first embodiment. The twenty-eighth embodiment also achieves the following effects [19-1], [19-2].

(249) [19-1] The second connecting portion **82** has the bottom portion **883**, the first side portion **881** and the second side portion **882**. The bottom portion **883** extends in the axial direction of the connecting rod **60**. The first side portion **881** extends from the bottom portion **883** in the direction orthogonal to the axial direction of the connecting rod **60**. The second side portion **882** extends in the direction orthogonal to the axial direction of the connecting rod **60** from one side of the bottom portion **883** opposite to the first side portion **881**. Also, the arm protrusion **632** is inserted into the space formed by the bottom portion **883**, the first side portion **881** and the second side portion **882**. Note that the axial direction of the connecting rod **60** corresponds to one direction. The direction orthogonal to the axial direction of the connecting rod **60** corresponds to the direction crossing the extending direction of the bottom portion **883**. The arm protrusion **632** corresponds to a rod protrusion.

(250) In such manner, the second connecting portion **82** hardly comes off from the connecting rod

60. Moreover, the second connecting portion **82** and the connecting rod **60** can be easily connected without providing an additional member. Therefore, the cost of a pedal device **1** is reducible by an amount of the cost of the additional member.

(251) [19-2] The length of the second side portion **882** is shorter than the length of the first side portion **881** in the direction orthogonal to the axial direction of the connecting rod **60**. Additionally, the second side portion **882** includes the second side slope **884**. The second side slope **884** is formed on one side of the second side **882** opposite to a side of the arm protrusion **632**, and is inclined with respect to the extending direction of the second side **882**. Thereby, the second side slope **884** guides the arm protrusion **632** into the space formed by the bottom portion **883**, the first side portion **881** and the second side portion **882**. The second side slope **884** corresponds to a rod slope.

(252) Since the length of the second side portion **882** is shorter than the length of the first side portion **881**, the arm protrusion **632** is easily insertable from a side of the second side portion **882** into the space formed by the bottom portion **883**, the first side portion **881** and the second side portion **882**. Further, since the arm protrusion **632** is guided into such space by the second side slope **884**, it becomes easy to insert the arm protrusion **632** into such space.

Twenty-Ninth Embodiment

(253) In the twenty-ninth embodiment, a housing **5** further has an inclined surface **500** and a wall portion **510**. Further, the shape of a first stopper **70** is different from that of the first embodiment. Moreover, a pedal device **1** further includes a protection plate **515**. The other configurations are similar to those of the first embodiment.

(254) As shown in FIGS. **36** to **38**, the inclined surface **500** is inclined downward toward the lower side of the vehicle from a left side wall **12** to a right side wall **13**. Further, a portion of the inclined surface **500** on the lower side of the vehicle is connected to the right side wall **13**, and a connection point **502** between the inclined surface **500** and the right side wall **13** is shifted forward of the vehicle relative to a shaft **41** and a rotating plate **43**. Also, the wall portion **510** is connected to the inclined surface **500**. Further, the wall portion **510** extends upward toward the upper side of the vehicle from a boundary portion with the inclined surface **500**. A first stopper **70** extends downward from a side plate portion **45**. Note that, in FIGS. **36** to **38**, illustration of a second housing **20** is omitted in order to avoid complication.

(255) The protection plate **515** corresponds to a protection portion, and protects a connecting rod **60** and a cover member **80**. Specifically, the protection plate **515**, extending from a pedal back surface **405** to a side of the housing **5**, at least partially overlaps with the connecting rod **60** and the cover member **80** in the extending direction of a rotation axis CL. Further, a part of the protection plate **515** and a part of the left side wall **12** of the housing **5** overlap in the extending direction of the rotation axis CL when a pedal **40** is not stepped on. Further, a part of the protection plate **515** and a part of the right side wall **13** of the housing **5** overlap in the extending direction of the rotation axis CL when the pedal **40** is not stepped on. Although the protection plate **515** is provided on both of the right side and the left side of the pedal back surface **405**, the protection plate **515** is not limited to such configuration, i.e., may be provided only on one of the right side or the left side, for example. Moreover, although the protection plate **515** is mentioned as a protection portion, the shape of the protection portion is not limited to be formed in a plate shape, and may also have, for example, a bar shape.

(256) Here, FIG. **39** shows a fully stepped-on state in which the pedal **40** is rotated to the farthest by a step-on operation of the driver of the vehicle. As shown in FIG. **39**, in the fully stepped-on state, a space between the pedal **40** and the housing **5** has a first region α and a second region β in a direction orthogonal to the pedal back surface **405**. Further, in the direction orthogonal to the pedal back surface **405**, a maximum distance from the pedal back surface **405** to the housing **5** in the second region β is longer than a maximum distance from the pedal back surface **405** to the housing **5** in the first region α . Also, the second region β is defined by the pedal back surface **405** and the

inclined surface **500**. Note that in FIG. **39**, the first region α and the second region β are respectively indicated by arrows. Also, in FIG. **39**, illustration of the second housing **20** is omitted in order to avoid complication.

(257) Also, the protection plate **515** does not cover the second region β in the space between the pedal **40** and the housing **5**. Therefore, the second region β is open on one side or the other side in the direction of the rotation axis CL. Thus, the inclined surface **500** of the housing **5** is not covered with the protection plate **515**, and is open to one side of the rotational axis CL direction where the protection plate **515** is arranged. In such manner, even if foreign matter is caught between the pedal back surface **405** and the housing **5**, when the driver of the vehicle steps on the pedal **40**, foreign matter pushed out to the outside of the housing **5** along the inclined surface **500**, without interfering with the protection plate **515**.

(258) The twenty-ninth embodiment is configured in the above-described manner. The twenty-ninth embodiment also achieves corresponding effects as the first embodiment. The twenty-ninth embodiment also achieves the following effects [20-1], [20-2].

(259) [20-1] Here, as described in Japanese Patent No. 6881257, there is a known vehicle pedal device including a comparative pedal pad, a comparative housing, a comparative support, and a comparative protection plate. The comparative protection plate is provided on the pedal pad, and is arranged outside in the vehicle width direction with respect to a space defined by the comparative pedal pad in a fully stepped-on state where the pedal is fully stepped on, the comparative housing, and the comparative support. Further, the comparative protection plate overlaps with the space. In such configuration, foreign matter intruding into the vehicle pedal device accumulates between the comparative pedal pad and the comparative housing, and the accumulated foreign matter is hard to be discharged because the comparative protection plate covers the comparative housing.

(260) On the other hand, the pedal device **1** further includes the protection plate **515**. By extending from the pedal back surface **405** toward the housing **5**, the protection plate **515** at least partially overlaps with the connecting rod **60** and the cover member **80** in the extending direction of the rotation axis CL. Further, the first region α and the second region β are formed at a position between the pedal back surface **405** and the housing **5** when the pedal **40** is rotated to the farthest by the step-on operation of the driver of the vehicle. Further, in the direction orthogonal to the pedal back surface **405**, the distance from the pedal back surface **405** to the housing **5** in the second region β is longer than the distance from the pedal back surface **405** to the housing **5** in the first region α . Further, since the protection plate **515** does not cover the second region β , the second region β is at least open to one side or other side in the extending direction of the rotation axis CL, i.e., open to the side where the protection plate **515** is arranged. Therefore, even if foreign matter is caught between the pedal back surface **405** and the housing **5**, the foreign matter is easily discharged to the outside of the pedal device **1** due to discharge from the second region β .

(261) [20-2] When the pedal **40** is not stepped on by the driver of the vehicle, a part of the protection plate **515** overlaps with a part of the left side wall **12** and the right side wall **13** of the housing **5** in the extending direction of the rotation axis CL.

(262) In such manner, even if a force acts on the protection plate **515** from the outside of the pedal device **1**, such as a case when the protection plate **515** is kicked by the foot of the driver of the vehicle, such a load applied to the protection plate **515** is received by the left side wall **12** and the right side wall **13** of housing **5**. Therefore, the connecting rod **60** and the cover member **80** are readily protected by the protection plate **515**.

Thirtieth Embodiment

(263) The thirtieth embodiment is different from the twenty-ninth embodiment in the form of a protection plate **515**. Other than the above, the thirtieth embodiment corresponds to the twenty-ninth embodiment.

(264) Specifically, as shown in FIG. **40**, the width of a protection plate **515** is wider than that of the twenty-ninth embodiment. In such manner, when a connecting rod **60** and a cover member **80** are

projected in the extending direction of a rotation axis CL, the projection of the connecting rod **60** and the cover member **80** entirely overlaps with the protection plate **515**. Note that a second housing **20** is omitted in FIG. **40** in order to avoid complication.

(265) Also in the thirtieth embodiment, since the protection plate **515** does not cover a second region β , the second region β is open to at least one of both (right and left) sides along the extending direction of the rotation axis CL, i.e., on one side in which the protection plate **515** is arranged. Therefore, even if foreign matter is caught between a pedal back surface **405** and a housing **5**, the foreign matter is easily discharged to the outside of a pedal device **1** due to discharge from the second region β .

(266) The thirtieth embodiment is configured in the above-described manner. Also in the thirtieth embodiment, corresponding effects as the twenty-ninth embodiment can be obtained.

Other Embodiments

(267) The present disclosure is not limited to the above-described embodiments, and the above-described embodiments are appropriately modifiable. Further, individual elements or features of a particular embodiment are not necessarily essential unless it is specifically stated that the elements or the features are essential in the foregoing description, or unless the elements or the features are obviously essential in principle.

(268) In the above-described embodiments, the pedal device **1** is used as a brake pedal in the brake-by-wire system **150** that controls the brakes of a vehicle. On the other hand, a pedal device **1** is not limited to be used as a brake pedal. The pedal device **1** may also be used, for example, as an accelerator pedal for accelerating a vehicle.

(269) In the above-described embodiments, the pedal device **1** is an organ-type pedal device **1**. On the other hand, a pedal device **1** is not limited to be the organ-type pedal device **1**. For example, the pedal device **1** may also be a suspension-type pedal device **1**. In the suspension-type pedal device **1**, a portion of a pedal **40** in the rear of the vehicle relative to a rotation axis CL rotates toward a floor **2** or a dash panel in the vehicle compartment in response to an increase in the stepping force of the driver applied to the pedal **40**.

(270) In the above-described embodiments, the reaction force generating mechanism **30** uses the plate spring **31**, the large-diameter coil spring **33**, and the small-diameter coil spring **34** as elastic members. On the other hand, the number of elastic members of a reaction force generating mechanism **30** is not limited to three. The number of elastic members of the reaction force generating mechanism **30** may be one or more.

(271) In the above-described embodiment, in the brake-by-wire system **150**, hydraulic pressure is generated in the brake fluid flowing through the brake circuit **120** by the master cylinder **126**. On the other hand, the hydraulic pressure is not limited to be generated in the brake fluid flowing through a brake circuit **120** by a master cylinder **126**. For example, hydraulic pressure may be generated in the brake fluid flowing through the brake circuit **120** by a hydraulic pump.

(272) In the second embodiment, the second connecting portion **82** is connected to the pedal back surface **405** by welding, adhesion, two-color molding, or the like. On the other hand, a second connecting portion **82** is not limited to be connected to a pedal back surface **405** by welding, adhesion, two-color molding, or the like. For example, as shown in FIG. **41**, a second connecting portion **82** may also be connected to a pedal back surface **405** by partially inserting the second connecting portion **82** into a space recessed from the pedal back surface **405**. Further, as shown in FIG. **42**, a second connecting portion **82** may also be connected to a pedal back surface **405** by inserting a pedal protrusion **406** protruding from the pedal back surface **405** into a space formed by a bottom portion **883**, a first side portion **881** and a second side portion **882** of the second connecting portion **82**. Further, as shown in FIG. **43**, a second connecting portion **82** may also be connected to a pedal back surface **405** by inserting a second fixing member **92** such as a screw, a pin, or a snap fit into a second connecting portion hole **862** of the second connecting portion **82** and a hole formed in a pedal back surface **405**. Further, as shown in FIG. **44**, a pedal back surface **405**

and a second connecting portion **82** may be connected by a sandwich structure that sandwiches the second connecting portion **82** in between (a) a second fixing member **92** such as a binding band, a retaining ring or the like and (b) a pedal protrusion **406**.

(273) In the twenty-ninth and thirtieth embodiments, a part of the protection plate **515** and a part of the side surface of the housing **5** overlap in the direction of the rotation axis CL in a state in which the pedal **40** is not stepped on, but the present disclosure is not limited to such a structure. For example, by increasing the rigidity of a protection plate **515** by, for example, providing an increased thickness for the protection plate **515**, a part of the protection plate **515** and a part of the side surface of a housing **5** do not have to overlap in the direction of a rotation axis CL.

(274) The embodiments described above may also be combined with each other when appropriate.

Claims

1. A pedal device comprising: a pedal configured to rotate about a rotation axis when stepped on by an operator; a connecting rod connected to a back surface of the pedal, opposite to a stepping surface stepped on by the operator, to rotate with the pedal about the rotation axis; a reaction force generating mechanism configured to generate a reaction force against a stepping force of the operator applied to the pedal by receiving a force from the pedal through the connecting rod and by an elastic deformation when the pedal is stepped on by the operator; a housing having a housing hole through which the connecting rod passes when the pedal is stepped on by the operator, and accommodating the reaction force generating mechanism; and a cover member including a housing connection portion connected to a housing end portion which is an end portion defining the housing hole in the housing, and a covering portion that is arranged at a position between the housing connection portion and the back surface of the pedal, and is connected to the housing connection portion, to cover the housing holes, the housing includes a first housing including the housing end portion, and a second housing, the first housing is provided with an opening at a position different from the housing hole, and a housing space communicating with the housing hole and the opening, the second housing is configured to close the opening of the first housing, a breathing hole is provided as a space between an opening-side end of the first housing and the second housing, the breathing hole is configured to discharge air in the housing space to an outside of the housing space after passing through the breathing hole, when the air in the housing space is compressed by receiving force from the pedal due to a stepping on of the pedal by the operator, and the breathing hole is configured to introduce air outside the housing space into the housing space through the breathing hole, when a force from the pedal is released due to stop of stepping on of the pedal subsequent to stepping on of the pedal by the operator to cause air in the housing space to expand.
2. The pedal device according to claim 1, wherein the covering portion has a bellows shape, and is configured to be elastically deformable and to generate a reaction force when the pedal is stepped on by the operator.
3. The pedal device according to claim 1, wherein the cover member further includes a pedal connecting portion connected to the covering portion and connected to the back surface of the pedal.
4. The pedal device according to claim 1, wherein the cover member further includes a mechanism connecting portion connected to a surface that contacts the connecting rod of the reaction force generating mechanism and connected to the covering portion.
5. The pedal device according to claim 1, wherein the cover member further includes a rod connecting portion connected to the covering portion and connected to the connecting rod.
6. The pedal device according to claim 5, wherein the connecting rod includes a rod recess recessed in a direction orthogonal to the axial direction of the connecting rod, the rod recess includes a rod bottom surface and a rod side surface connected to the rod bottom surface, and the rod connecting portion includes an insertion portion inserted into a space defined by the rod bottom surface and the

rod side surface, and a protruding portion protruding from the insertion portion in a direction orthogonal to the axial direction of the connecting rod.

7. The pedal device according to claim 5, wherein the connecting rod includes a rod protrusion protruding from the side surface of the connecting rod in a direction orthogonal to the axial direction of the connecting rod, the rod connecting portion includes a bottom portion extending in an extending direction, a first side portion extending from the bottom portion in a direction crossing the extending direction of the bottom portion, and a second side portion extending from the bottom portion on a side opposite to the first side portion, in a direction crossing the extending direction of the bottom portion, and the rod protrusion is inserted into a space defined by the bottom portion, the first side portion and the second side portion.

8. The pedal device according to claim 7, wherein a length of the second side portion is shorter than a length of the first side portion, the second side portion includes a rod slope formed on an opposite side of the rod protrusion, and the rod slope is inclined with respect to an extending direction of the second side portion to guide the rod protrusion into a space defined by the bottom portion, the first side portion, and the second side portion.

9. The pedal device according to claim 1, wherein the housing connection portion includes a first extension portion extending in one direction, a second extension portion extending from the first extension portion in an extending direction crossing the one direction of the first extension portion, and a third extension portion extending, from one side of the second extension portion opposite to the first extension portion, in a direction crossing the extending direction of the second extension portion, and the housing end portion is inserted into a space defined by the first extension portion, the second extension portion and the third extension portion.

10. The pedal device according to claim 9, wherein a length of the third extension portion is shorter than a length of the first extension portion, the third extension portion includes a housing slope provided at a side opposite the housing end, and the housing slope is inclined with respect to an extending direction of the third extension portion to guide the housing end portion into a space that is defined by the first extension portion, the second extension portion and the third extension portion.

11. The pedal device according to claim 1, wherein the housing end portion includes an end recess recessed therefrom, the end recess includes an end bottom surface and an end side surface connected to the end bottom surface, and the housing connection portion is inserted into a space defined by the end bottom surface and the end side surface.

12. The pedal device according to claim 1, wherein the housing connection portion includes a low-strength portion inserted into the housing hole and connected to the housing end portion, and a high-strength portion covered by the low-strength portion and having a Young's modulus higher than that of the low strength portion.

13. The pedal device according to claim 1, wherein the opening of the first housing is provided at a position opposite to the housing hole.

14. The pedal device according to claim 1, wherein the housing is provided with a housing space communicating with the housing hole, and a filter hole communicating with an outside of the housing and the housing space, the pedal device further comprising a filter configured to allow air to pass therethrough without passing foreign matter flowing from the outside of the housing through the filter hole toward the space inside the housing, when the pedal is stepped on by the operator to cause compression of air in the housing space by force received from the pedal, air in the housing space passes through the filter hole and the filter to discharge to the outside of the housing, and when the operator stops step-on operation of the pedal in a step-on state of the pedal to release force from the pedal and to cause expansion of air in the housing space, air from the outside of the housing space is introduced into the housing space through the filter hole and the filter.

15. A pedal device comprising: a pedal configured to rotate about a rotation axis when stepped on

by an operator; a connecting rod connected to a back surface of the pedal, opposite to a stepping surface stepped on by the operator, to rotate with the pedal about the rotation axis; a reaction force generating mechanism configured to generate a reaction force against a stepping force of the operator applied to the pedal by receiving a force from the pedal through the connecting rod and by an elastic deformation when the pedal is stepped on by the operator; a housing having a housing hole through which the connecting rod passes when the pedal is stepped on by the operator, and accommodating the reaction force generating mechanism; a cover member including a housing connection portion connected to a housing end portion which is an end portion defining the housing hole in the housing, and a covering portion that is arranged at a position between the housing connection portion and the back surface of the pedal, and is connected to the housing connection portion, to cover the housing hole; and a protection portion extending from the back surface of the pedal toward the housing to partially overlap with at least a part of the connecting rod and the cover member in an extending direction of the rotation axis, wherein a first region and a second region are provided at a position between the back surface of the pedal and the housing when the pedal is rotated to a farthest by the step-on operation of the operator, in a direction orthogonal to the pedal back surface, a distance from the back surface of the pedal to the housing in the second region is longer than a distance from the back surface of the pedal to the housing in the first region, and the protection portion is configured without covering the second region, to open the second region toward at least one direction of an extension line of the rotation axis, on which the protection portion is arranged.

16. The pedal device according to claim 15, wherein a part of the protection portion overlaps with a part of the housing in an extending direction of the rotation axis when the pedal is not stepped on by the operator.
